Aufgabe 7.1.3: Kamerakalibrierung

Ausführung des Matlab-Skripts, welches die Kameraparameter erzeug (durch Exportieren der Kamaeraparameter erzeugt):

calibration_camera

Standard Errors of Estimated Camera Parameters

Intrinsics

Focal length (pixels): [1110.7087 +/- 2.5894 1114.9353 +/- 2.4828]

Principal point (pixels):[781.7992 +/- 1.0573 584.7259 +/- 1.5142]

Radial distortion: [0.1447 +/- 0.0099 -0.3674 +/- 0.0479]

Extrinsics

Rotation vectors:

[0.0654 +/- 0.0015 -0.8999 +/- 0.0011 -0.2643 +/- 0.0006]

[0.1600 +/- 0.0012 0.1234 +/- 0.0010 0.0287 +/- 0.0002]

[0.3831 +/- 0.0011 -0.1818 +/- 0.0009 0.0223 +/- 0.0002]

Translation vectors (millimeters):

[-78.3968 +/- 0.2800 -15.4188 +/- 0.3937 293.5938 +/- 0.7547]

[-75.4183 +/- 0.3052 -55.6406 +/- 0.4538 328.3644 +/- 0.7904]

[-85.5477 +/- 0.2739 -57.6728 +/- 0.3984 281.3114 +/- 0.7278]

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Dokumentation:

Die Kamerakalibrierung liefert mit einem mittleren Reprojektionsfehler von 0,36 Pixeln (im Bild mit dem cameraParams Objekt gut sichtbar) eine gute Genauigkeit. Die bestimmten intrinsischen Parameter beschreiben Brennweite, Hauptpunkt und radiale Verzeichnung der Kamera. (durch das Skript oben sichtbar) Die unterschiedlichen Extrinsikparameter zeigen die verschiedenen Positionen und Orientierungen der Kamera bei den Aufnahmen.

Ausführung des Matlab-Skripts, welches die Stereoparameter erzeug (durch Exportieren der Parameter erzeugt):

stereo_calibration

Standard Errors of Estimated Stereo Camera Parameters

Camera 1 Intrinsics

Focal length (pixels): [1038.0283 +/- 0.6533 1037.9142 +/- 0.6389]
Principal point (pixels):[656.0836 +/- 0.3408 485.5483 +/- 0.2639]
Radial distortion: [-0.3617 +/- 0.0008 0.1866 +/- 0.0026]

Camera 1 Extrinsics

Rotation vectors:

[0.1680 +/- 0.0007	-0.0271 +/- 0.0006	3.1125 +/- 0.0001]
[0.1995 +/- 0.0006	-0.0523 +/- 0.0005	-3.0991 +/- 0.0000]
[0.4187 +/- 0.0005	-0.1061 +/- 0.0004	-3.1113 +/- 0.0001]
[0.5239 +/- 0.0005	-0.0604 +/- 0.0004	-3.0552 +/- 0.0001]
[0.6807 +/- 0.0006	-0.0306 +/- 0.0005	-3.0331 +/- 0.0001]
[0.3513 +/- 0.0007	-0.0993 +/- 0.0006	-3.0334 +/- 0.0001]
[0.0212 +/- 0.0007	-0.1179 +/- 0.0007	-3.0833 +/- 0.0000]
[-0.2765 +/- 0.0008	-0.0848 +/- 0.0007	-3.0943 +/- 0.0001]
[-0.4407 +/- 0.0007	-0.1119 +/- 0.0006	-3.0652 +/- 0.0001]
[-0.2537 +/- 0.0008	-0.1334 +/- 0.0007	-3.1039 +/- 0.0001]

Translation vectors (millimeters):

[163.9861 +/- 0.1137	52.5579 +/- 0.0926	345.5709 +/- 0.2344]
[85.2874 +/- 0.1210	44.3797 +/- 0.0948	368.0356 +/- 0.2312]
[52.4009 +/- 0.1197	44.2461 +/- 0.0933	365.3883 +/- 0.2217]
[11.4672 +/- 0.1200	45.5439 +/- 0.0933	365.8657 +/- 0.2197]
[-39.9072 +/- 0.1621	34.9517 +/- 0.1251	490.5845 +/- 0.2901]
[2.4951 +/- 0.1570	40.9442 +/- 0.1221	478.4337 +/- 0.2988]
[68.3993 +/- 0.1532	38.8583 +/- 0.1194	465.4561 +/- 0.2949]
[142.1840 +/- 0.1495	38.4727 +/- 0.1193	455.5970 +/- 0.2945]
[177.5501 +/- 0.1413	38.3189 +/- 0.1155	432.4844 +/- 0.2869]
[220.7904 +/- 0.1698	-3.4254 +/- 0.1398	522.1333 +/- 0.3497]

Camera 2 Intrinsics

Focal length (pixels): [1042.4813 +/- 0.6644 1042.2688 +/- 0.6534]
Principal point (pixels):[640.5981 +/- 0.3305 479.0651 +/- 0.2633]
Radial distortion: [-0.3614 +/- 0.0007 0.1822 +/- 0.0022]

Position And Orientation of Camera 2 Relative to Camera 1

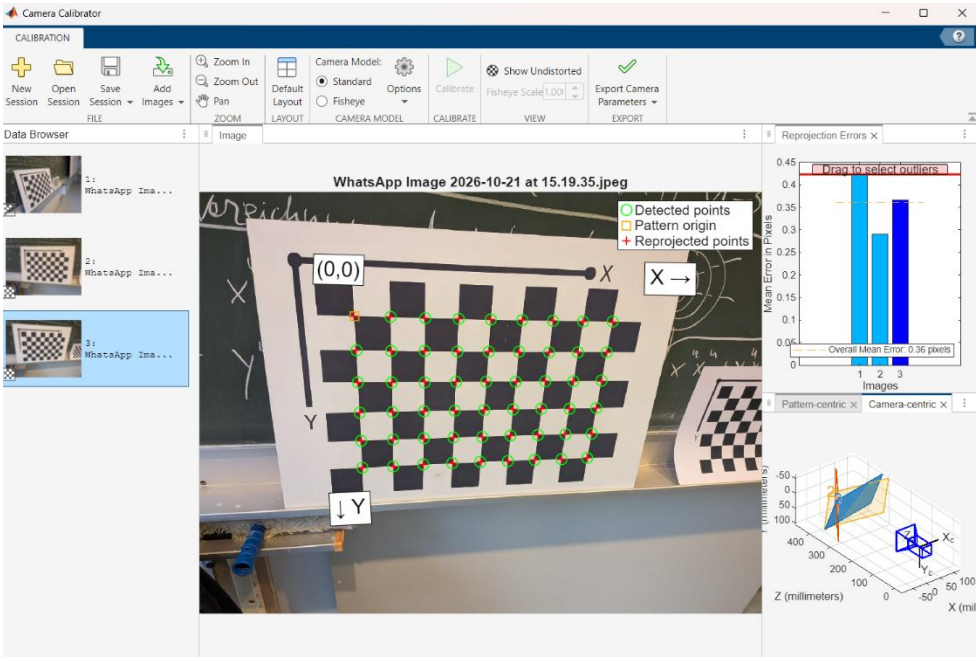
Rotation of camera 2: [-0.0037 +/- 0.0002 0.0050 +/- 0.0004 -0.0002 +/- 0.0000]
Translation of camera 2 (millimeters):[-27.7481 +/- 0.0093 -0.0927 +/- 0.0096 -0.0060 +/- 0.0405]

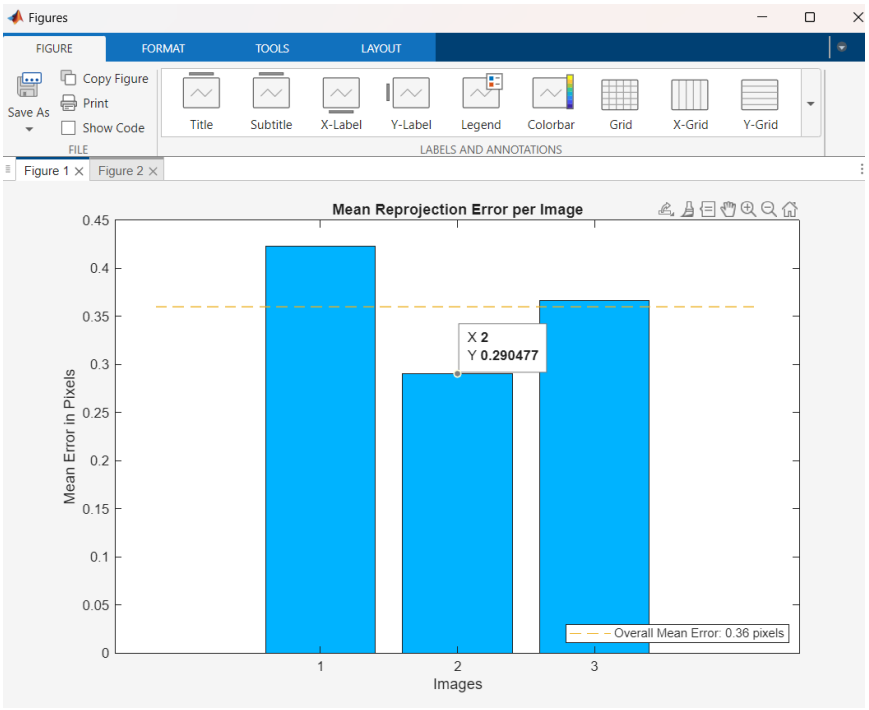
>>

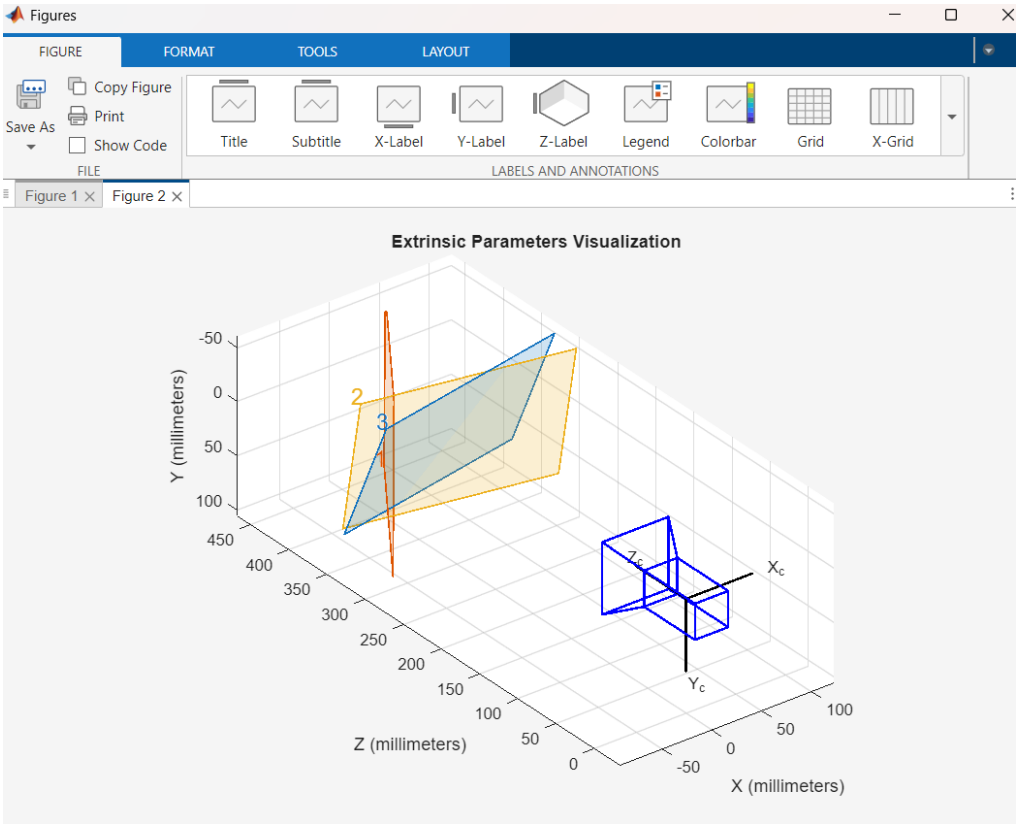
Interpretation:

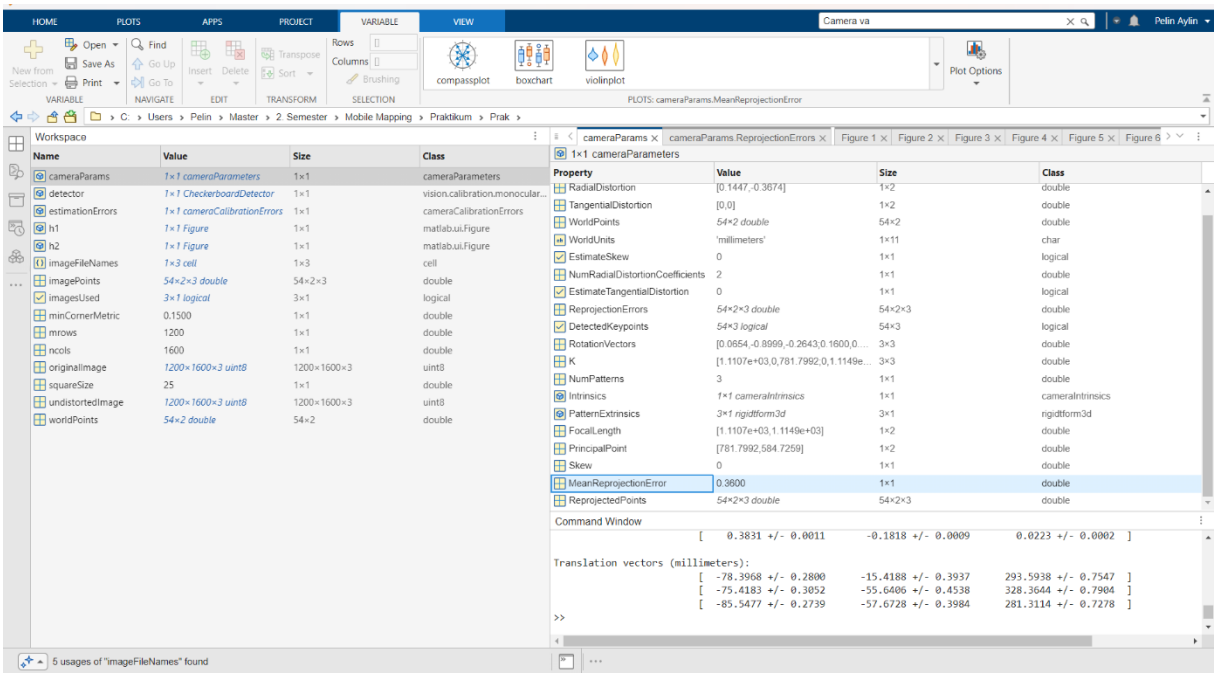
Die Stereo-Kalibrierung zeigt einen mittleren Reprojektionsfehler von etwa 0,06 Pixeln und damit eine sehr gute Genauigkeit (im Bild mit dem stereoParams Objekt gut sichtbar). Die intrinsischen Parameter beider Kameras wurden bestimmt und die Extrinsikparameter beschreiben ihre relative Lage und Orientierung. Insgesamt sind die Ergebnisse für die weitere Stereo-Bildverarbeitung und Bildentzerrung gut geeignet.

Kalibrierung eins Monosystems (mit eigenen Bildern)

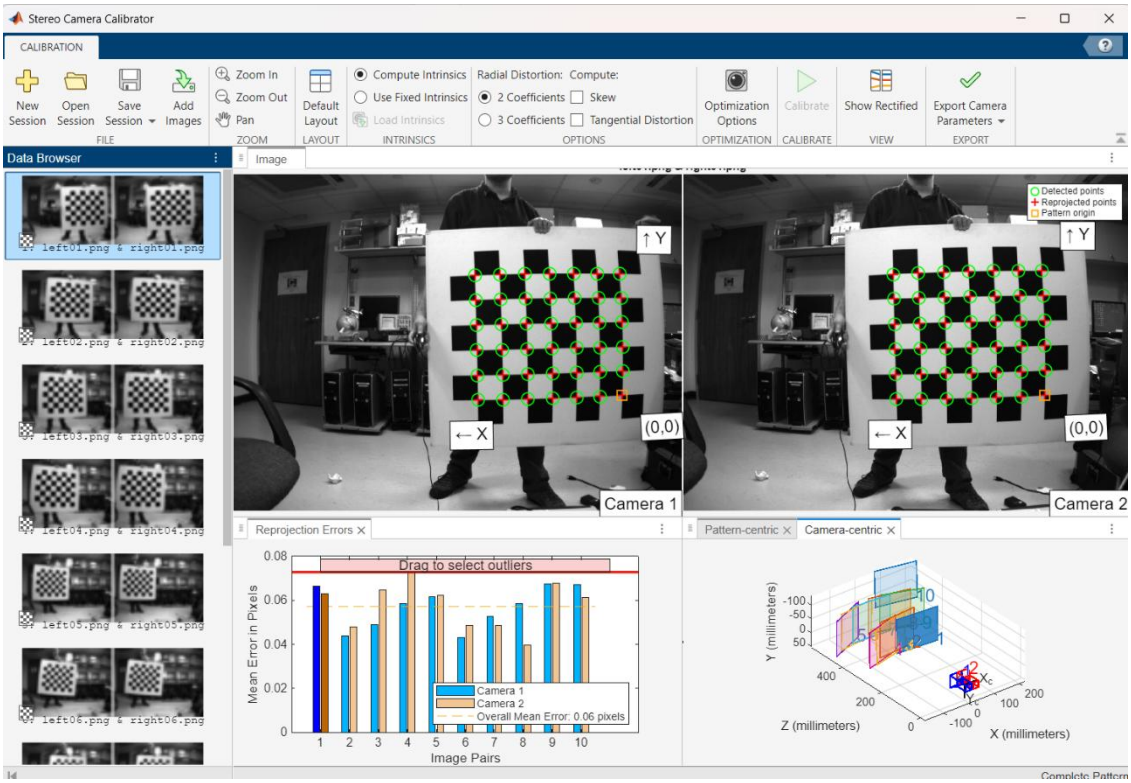


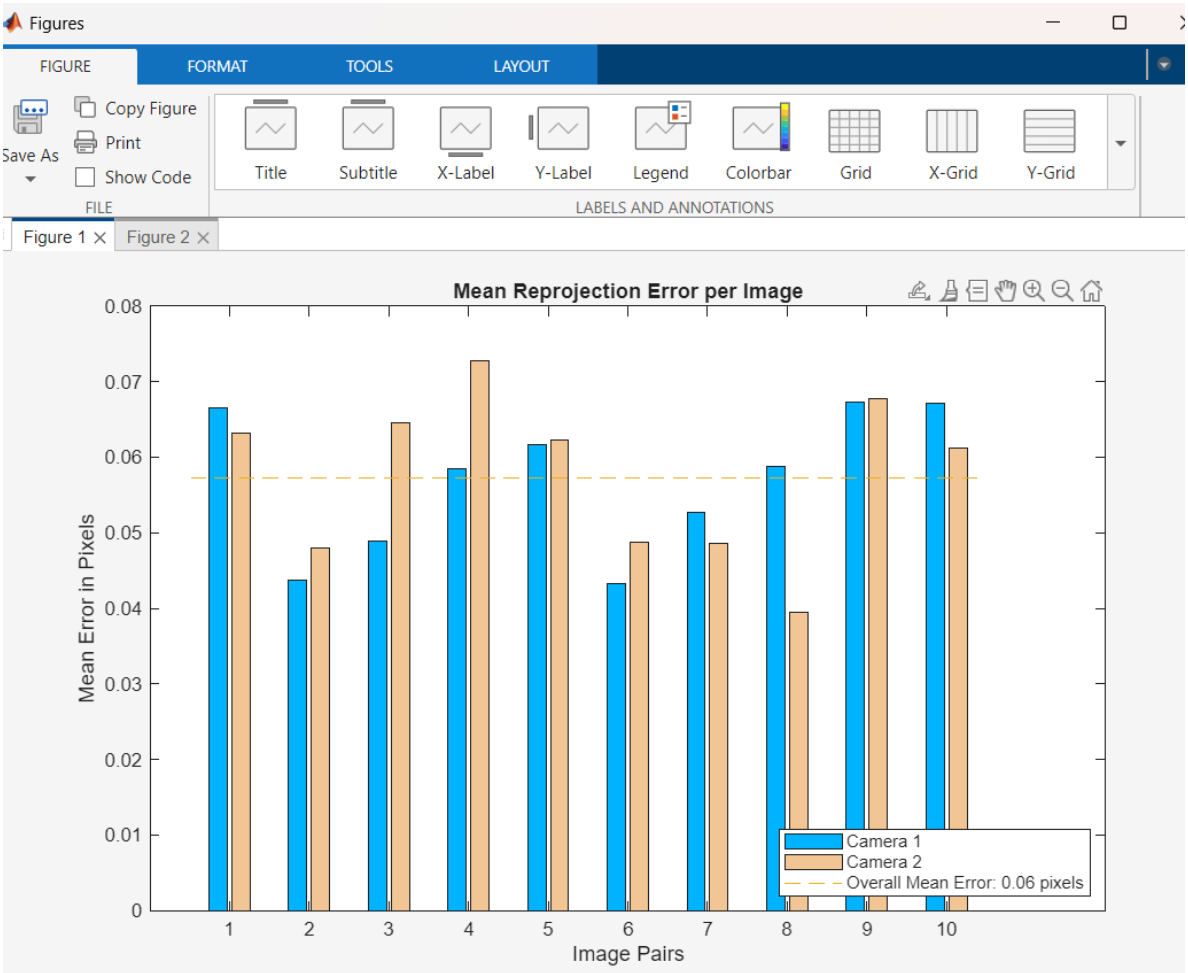


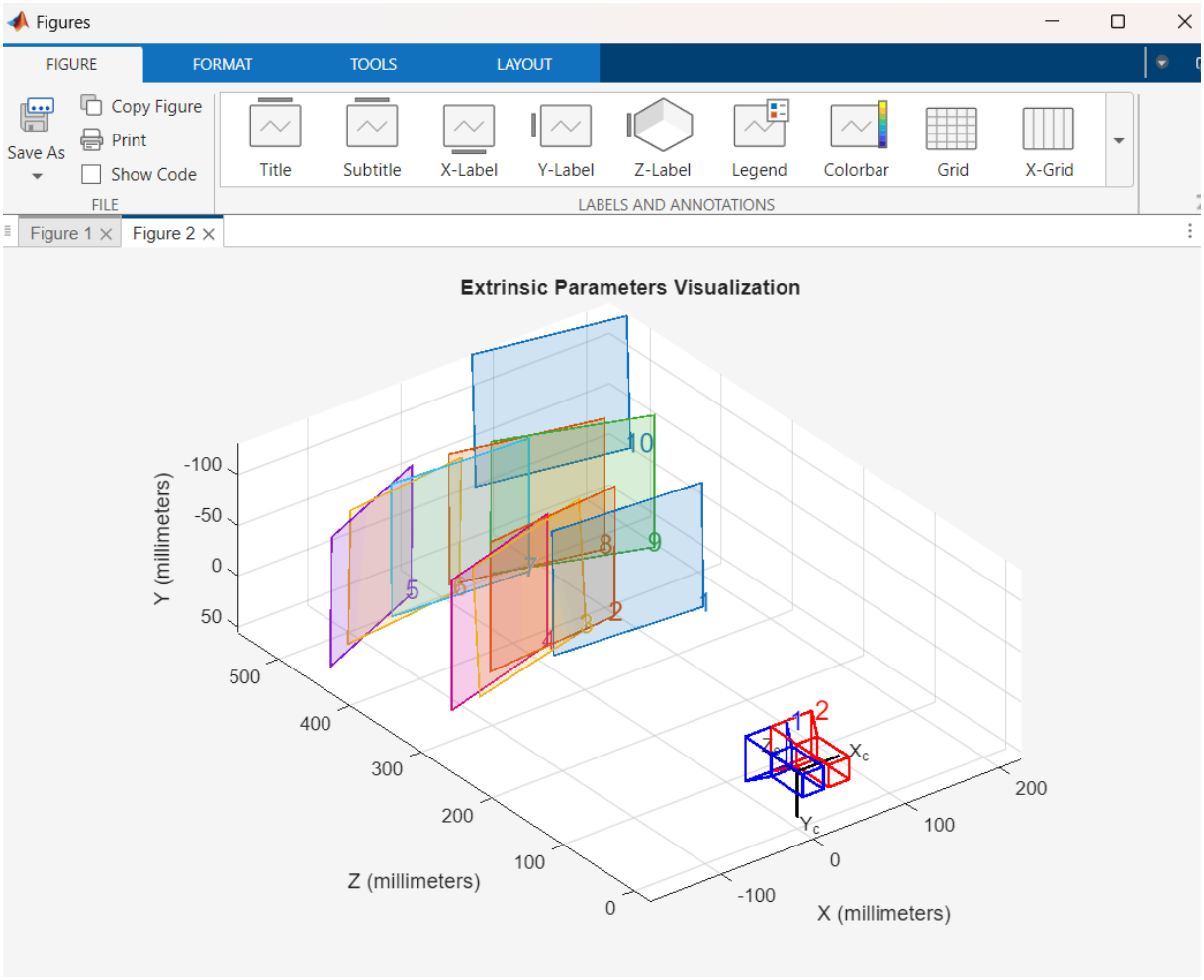

Figures
FIGURE
FORMAT
TOOLS
LAYOUT
Save As
Copy Figure
Print
Show Code
FILE
Title
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
LABELS AND ANNOTATIONS
Figure 1
Figure 2
Extrinsic Parameters Visualization
Y (millimeters)
Z (millimeters)
X (millimeters)
Xc
Yc
Zc

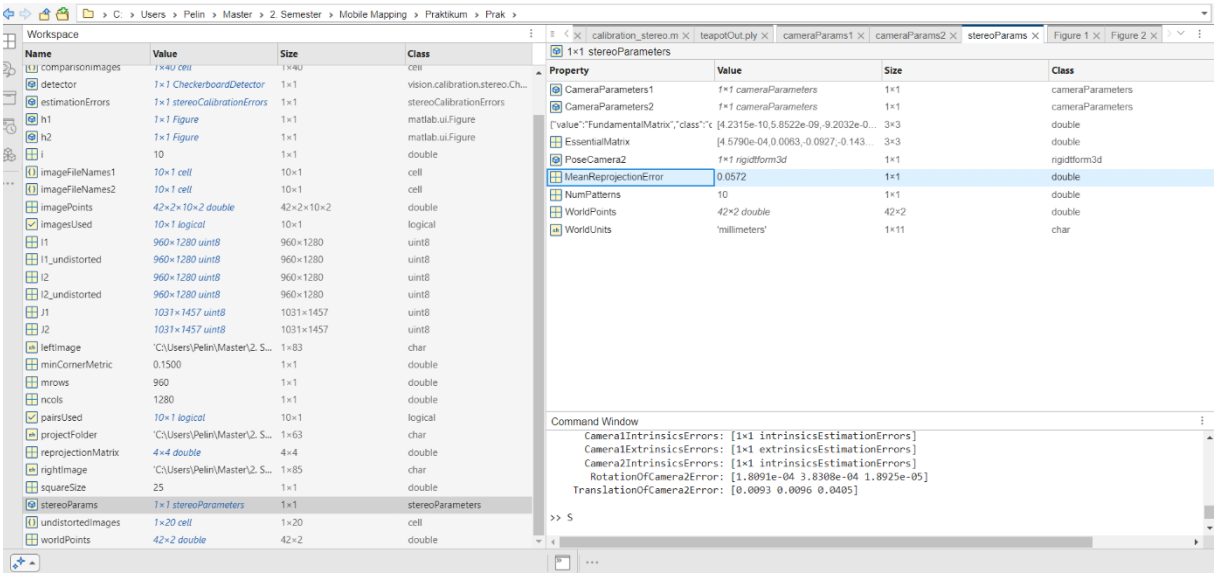

HOME
PLOTS
APPS
PROJECT
VARIABLE
VIEW
compassplot
boxchart
violinplot
Plot Options
Workspace
cameraParams
cameraParams ReprojectionErrors
1×1 cameraParameters
Command Window
Translation vectors (millimeters):
5 usages of "imageFileNames" found

Stereokalibrierung (mit Bildern aus der Toolbox)






Figures
FIGURE
FORMAT
TOOLS
LAYOUT
Save As
Copy Figure
Print
Show Code
FILE
Title
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
LABELS AND ANNOTATIONS
Figure 1
Figure 2
Extrinsic Parameters Visualization
Y (millimeters)
-100
-50
0
50
Z (millimeters)
500
400
300
200
100
0
X (millimeters)
200
100
0
-100
Xc
Yc
Zc
1
2
3
4
5
6
7
8
9
10


Workspace
Name
Value
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1×1 CheckerboardDetector
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estimationErrors
1×1 stereoCalibrationErrors
stereoCalibrationErrors
h1
1×1 Figure
matlab.ui.Figure
h2
1×1 Figure
matlab.ui.Figure
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double
imageFileNames1
10×1 cell
cell
imageFileNames2
10×1 cell
cell
imagePoints
42×2×10×2 double
double
imagesUsed
10×1 logical
logical
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uint8
I1_undistorted
960×1280 uint8
uint8
I2
960×1280 uint8
uint8
I2_undistorted
960×1280 uint8
uint8
J1
1031×1457 uint8
uint8
J2
1031×1457 uint8
uint8
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double
mrows
960
double
ncols
1280
double
pairsUsed
10×1 logical
logical
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4×4 double
double
rightImage
'C:\Users\Pelin\Master\2. S...
char
squareSize
25
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stereoParams
1×1 stereoParameters
stereoParameters
undistortedImages
1×20 cell
cell
worldPoints
42×2 double
double
1×1 stereoParameters
Property
Value
Size
Class
CameraParameters1
1×1 cameraParameters
CameraParameters2
1×1 cameraParameters
EssentialMatrix
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PoseCamera2
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MeanReprojectionError
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NumPatterns
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WorldPoints
42×2 double
WorldUnits
'millimeters'
Command Window
Camera1IntrinsicsErrors: [1×1 intrinsicsEstimationErrors]
Camera1ExtrinsicsErrors: [1×1 extrinsicsEstimationErrors]
Camera2IntrinsicsErrors: [1×1 intrinsicsEstimationErrors]
RotationOfCamera2Error: [1.8091e-04 3.8308e-04 1.8925e-05]
TranslationOfCamera2Error: [0.0093 0.0096 0.0405]
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